

Problem Set 2

Prof. Oke

CEE 260/MIE 273: Probability & Statistics in Civil Engineering

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*Due Tuesday, September 22, 2026 at 12:59 PM as PDF uploaded via Gradescope. If it helps and if possible, you can write your responses directly on this document and upload it instead. **Show as much work as possible in order to get FULL credit.** There are 3 problems with a total of 20 points available.*

Problem 1 (5 points)

Respond “T” (*True*) or “F” (*False*) to the following statements. Use the boxes provided. Each response is worth 1 point.

(i)

☐

An impossible event has a probability less than or equal to 0.

(ii)

☐

A certain event is equivalent to the sample space of outcomes.

(iii)

☐

If A and B are mutually exclusive, then $P(AB) \neq 0$.

(iv)

☐

Given an event E , then the probability of its complement can be given as $P(\overline{E}) = 1 + P(E)$

(v)

☐

A and B are collectively exhaustive events. If $P(B) = 0.4$, then $P(A) = 0.6$.

Problem 2 (7 points)

Show your work/explanations in order to receive full credit. Single number answers are not sufficient.

- [2] (a) License plates in a certain state consist of 3 letters followed by 3 digits. How many different license plates can be manufactured?
- [2] (b) Your computer password must have only 6 characters. Each character can be a lowercase letter (e.g. a), an uppercase letter (e.g. A) or a digit (e.g. 0). How many valid passwords could you possibly create?
- [3] (c) Your operating system is tightening its security protocols and asks you to update your password. It now requires that in addition to the earlier rules, you must now have at least one uppercase letter in your computer password. How many valid passwords could you possibly now create?

Problem 3 (8 points)

Data collected at elementary schools in DeKalb County, GA suggest that each year roughly 25% of students miss exactly one day of school, 15% miss 2 days, and 30% miss 3 or more days due to sickness.

Example

Let X be the number of school days missed in a year.

The probability that a student chosen at random misses 2 or more days of school in a year is given by:

$$P(X \geq 2) = P(X = 2) + P(X \geq 3) = 0.15 + 0.30 = 0.45$$

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- (a) What is the probability that a student chosen at random does not miss any days of school due to sickness this year?
- (b) What is the probability that a student chosen at random misses no more than one day?
- (c) What is the probability that a student chosen at random misses at least one day?
- (d) What is the probability that a student chosen at random misses one or two days of school a year?